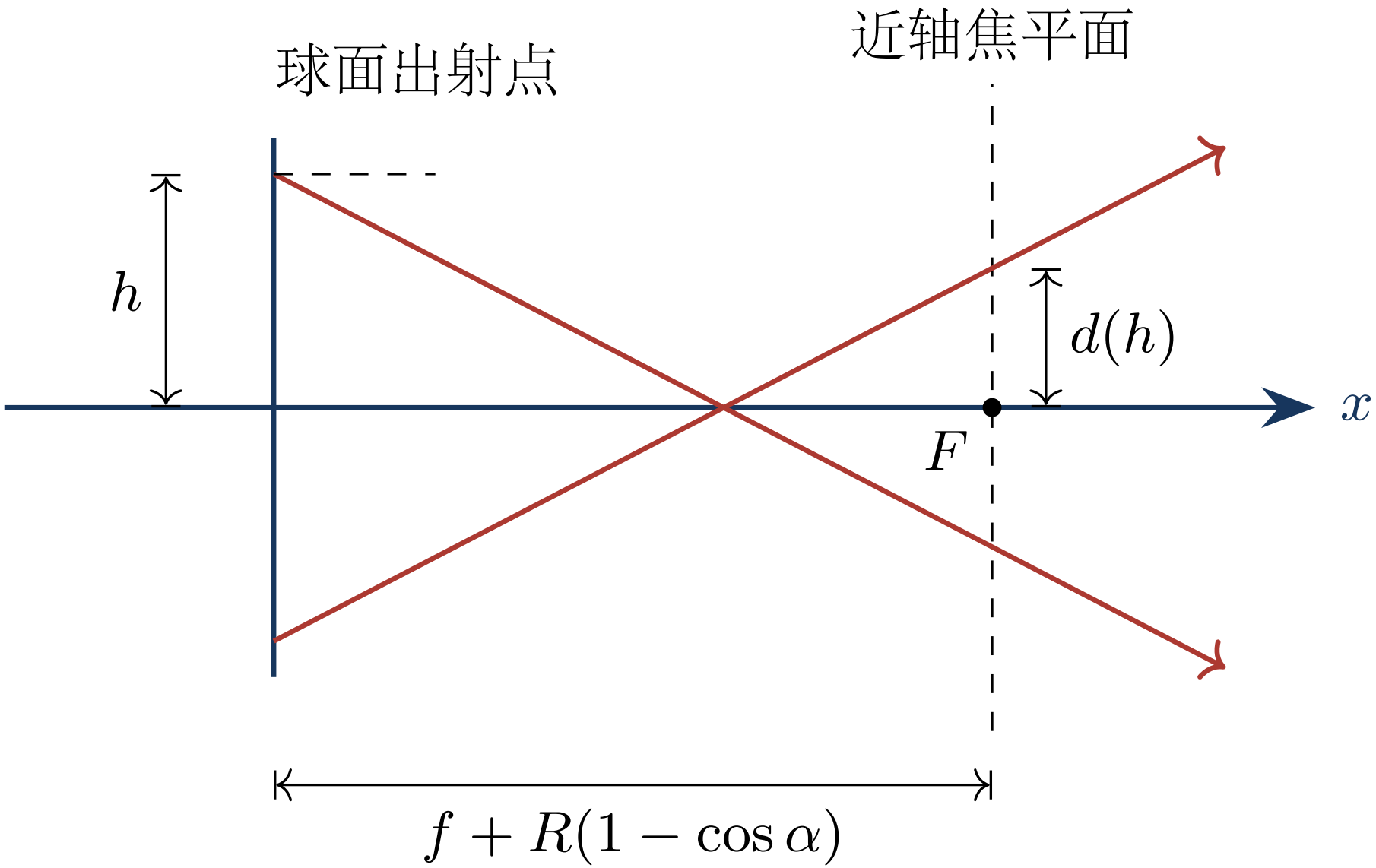

$$d = [f + R(1 - \cos\alpha)] \tan\beta - h$$